

Bessel Functions of the First Kind

Power Series Definition

Motivation

Bessel functions arise naturally when solving PDEs with radial symmetry:

- The wave equation on a circular membrane
- The heat equation in cylindrical coordinates
- Laplace's equation in polar coordinates

Bessel functions are defined by an ODE or a power series.

Bessel's Equation

For order $\nu \in \mathbb{R}$, Bessel's equation is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

This is a second-order linear ODE with a regular singular point at $x = 0$.

We seek a solution of the form

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0.$$

- This is the *Frobenius method*
- r is determined by the indicial equation

Substituting the Frobenius Series

Assume a Frobenius expansion

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0.$$

Then

$$y'(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1},$$

$$y''(x) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-2}.$$

Multiply by the coefficients in Bessel's equation:

$$x^2 y'' + x y' + (x^2 - \nu^2) y.$$

Collecting Like Powers of x

After substitution, we obtain

$$\begin{aligned} \sum_{k=0}^{\infty} a_k(k+r)(k+r-1)x^{k+r} + \sum_{k=0}^{\infty} a_k(k+r)x^{k+r} \\ + \sum_{k=0}^{\infty} a_k x^{k+r+2} - \nu^2 \sum_{k=0}^{\infty} a_k x^{k+r} = 0. \end{aligned}$$

Shift indices in the x^{k+r+2} term so that all sums involve x^{k+r} .

The Indicial Equation

The lowest power of x that appears is x^r .

Its coefficient is

$$a_0 \left[(r)(r-1) + r - \nu^2 \right] = a_0 (r^2 - \nu^2).$$

Since $a_0 \neq 0$, we must have

$$r^2 - \nu^2 = 0.$$

Indicial roots:

$$r = \pm \nu.$$

Choosing the Exponent r

From the indicial equation

$$r^2 - \nu^2 = 0,$$

we take

$$r = \nu$$

to obtain the solution regular at the origin.

With $r = \nu$, the recurrence simplifies to

$$a_{k+2} = -\frac{1}{(k + \nu + 2)^2 - \nu^2} a_k = -\frac{1}{(k + 2)(k + 2\nu + 2)} a_k.$$

This is a two-step recurrence relation.

Role of Initial Coefficients

Because the recurrence advances by two indices:

- a_0 determines all *even* coefficients
- a_1 determines all *odd* coefficients

Hence the general solution is a linear combination of:

- an even series
- an odd series

These correspond to two linearly independent solutions.

Selecting the J_ν Solution

Choose the indicial root $r = \nu$.

Then:

- Setting $a_1 = 0$ produces a solution bounded at $x = 0$
- This yields the Bessel function J_ν

The normalization constant a_0 is chosen for convenience.

Power Series Definition of J_ν

The *Bessel function of the first kind* of order ν is defined by

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}$$

- Converges for all $x \in \mathbb{R}$
- Entire function of x

Special Case: Integer Order

For $n \in \mathbb{N} \cup \{0\}$,

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{2k+n}.$$

- Most common case in PDE applications
- No Gamma function needed

Order 0:

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^4 2!} - \dots$$

Order 1:

$$J_1(x) = \frac{x}{2} - \frac{x^3}{2^3 1!} + \frac{x^5}{2^5 2!} - \dots$$

Behavior Near the Origin

From the series definition:

$$J_\nu(x) \sim \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu \quad \text{as } x \rightarrow 0.$$

- Finite at $x = 0$ for $\nu \geq 0$
- Explains selection of J_ν in physical problems

Why J_ν Is Preferred

- Regular (bounded) at the origin
- Orthogonal under suitable boundary conditions
- Natural eigenfunctions for radial Sturm–Liouville problems

The second solution (Y_ν) is singular at $x = 0$.

Summary

- Bessel's equation admits series solutions via Frobenius
- J_ν is defined by a globally convergent power series
- Integer-order Bessel functions dominate PDE applications

Power series $\Rightarrow$ structure, regularity, and applications